Write-up

An analyst co-pilot that clarifies, checks definitions, picks the right data layer and shows its evidence.

01 What I built and why

1 Clarify — metric, period, population; asks if ambiguous

2 Verify definition — metric exists and its definition in registry + catalog

3 Route to a layer — Gold if it fits; Silver for detail; Bronze for raw audit

4 Query + evidence — read-only, bounded; one log per query, never overwritten

5 Test hypotheses — ≤3 per round, ≤3 rounds; stop when explained

6 Report — pattern ≠ cause; fixed PDF template, ≤10 pages

Read by every run, in Claude Code or Codex: AGENTS.md · SKILL.md · catalog · registry

Guardrails: read-only authorizer · row and time limits

Gold — 8 monthly tables · fast standard answers

Silver — 10 cleaned tables · detail for investigations

Bronze — 20 views over raw tables · audits

One SQLite database; layers built in one transaction with 122 reconciliation checks. Step 3 decides which layer each query reads.

- **Clarifies instead of assuming.** When the metric, period or meaning is open, it asks first and states every assumption it keeps.
- **Verifies the metric before computing it.** It confirms the metric exists and uses its approved definition and SQL from the metric registry, plus joins and known gaps from the data catalog. Unsupported requests are declined, not improvised.
- **Maps the question to a layer for speed.** Gold answers standard monthly questions directly; Silver keeps per-user detail; Bronze is only for auditing raw values. Fewer, smaller queries keep answers fast.
- **Evidence for every number and LLM-neutral by design.** Each query is logged with its SQL, layer and metric version. Definitions, rules and guardrails live in files, not prompts, so Claude Code and Codex behave the same and a lesson becomes a rule for every later run.

02 How it performed: 18 blind test questions, two rounds

	Q01	Q02	Q03	Q04	Q05	Q06	Q07	Q08	Q09	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18
Round 1	Pass	Partial	Pass	Pass	Pass	Pass	Fail	Pass	Pass	Pass	Partial	Pass	Pass	Pass	Pass	Pass	Pass	Pass
Round 2	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Partial	Pass	Pass	Pass	Pass	Pass	Pass	Pass

Round 1: 15 pass · 1 partial · 2 fail **Round 2: 17 · 1 · 0**

Question type	Questions	Result	What happened
Simple numbers and filters	Q01 Q05 Q08 Q17 Q18	**All pass**	Exact to the user and the cent
Vague or ambiguous questions	Q03 Q10 Q15 Q18	**All pass**	Asked before choosing a metric, period or meaning
Unanswerable question	Q16	**Pass**	Declined and named the data that's missing
Data-quality traps	Q12 Q13 Q14	**All pass**	Caught both planted issues; didn't claim to have detected a reviewed one
Comparisons and lifecycle	Q04 Q06 Q09	**All pass**	Right time basis; groups reconcile to the total
Cohort conversion	Q07	**Fail → Pass**	Used April, the month the subscriber base forms, as a normal baseline. A new skill rule fixed it
Funnel drop-off	Q11	**Partial**	Right answer and gaps, but read April's start-up as a funnel decline; the new rule didn't trigger here
GM investigation	Q02	**Fail → Pass**	v1: headline numbers and story right, but some conclusions stated more firmly than the evidence allowed. After review, three certainty rules joined the skill and the agent's v2 passed; the issue didn't recur in 17 later runs

Scored against values computed beforehand from the raw tables. Correct numbers alone don't pass: a silent choice of definition, a missing caveat or an overstated conclusion fails. Round 2: Q02 is the agent's v2 after review; Q07 and Q11 were rerun, with Q01 and Q09 to check nothing broke; other results carry over. **Hardest:** comparisons that need context the data doesn't announce, such as a start-up month.

03 Running it for a 20-person product team

Time to answer (observed)	**Simple** (MAU by month): 2–3 min	**Normal** (refund impact): 5–6 min	**Complex** (MAU up, revenue flat): 10–12 min

- **Set-up for real data:** first build a data catalog and metric registry for the team's warehouse; the agent relies on both.
- **Cost (estimate):** running the 18 questions twice used about half of one 5-hour window on an INR 1,999-a-month Codex plan, so one seat covers a heavy day of questions: about INR 1,999 per active analyst a month. Token cost per run wasn't measured.
- **Lower cost and latency:** answer from the smallest layer that fits, with targeted queries instead of raw scans.
- **Guardrails:** data is only ever read: a read-only database role on a replica plus the query tool's read-only check, row and time limits; every figure traceable to its evidence log; sensitive columns kept out of the catalog.
- **Freshness:** real data keeps arriving, unlike this static dataset. Add a freshness tracker (last load per source and layer) that the agent checks first, warning or stopping when data is stale.
- **Quality over time:** rerun the 18 questions on every change to the skill, registry or data; track pass rate, clarification rate and query errors; review a weekly sample of real questions and turn failures into rules.

04 With more time

- Run the same questions in both runtimes for a direct Claude vs Codex comparison, plus held-out questions.
- Log tokens and time per run to replace the cost estimate with a measured cost per question.
- A short-answer mode for simple questions; a helper that drafts the catalog for a new database; a Slack or web front end.

Note on tools. Claude Code (Claude Opus 5.5, medium) built the framework, ran the GM investigation and computed the answer key. Codex (GPT Terra 6.1, medium) ran the evaluation, so the agent was tested in a runtime that didn't build it. ChatGPT (Terra 5.6, later 6.1, medium) reviewed the GM report and evaluation decisions independently.